

from January to March 2020. Thirty-eight oocytes, identified as immature after cumulus-oocyte complexes (COC) stripping for ICSI, were placed in our in vitro maturation medium with the addition of autologous cumulus cells. Oocytes that had reached the metaphase II stage after a maximum of 36 hours of maturation were microinjected. The fertilization and embryonic development potential of the in vitro matured oocytes were compared to those of 148 in vivo matured "siblings" oocytes from the same oocyte retrieval, and then also compared to those of 127 in vivo matured oocytes from different patients (control group). No significant difference was found in the main and secondary criteria of the study compared to the "siblings" in vivo matured oocytes from the same oocyte retrieval. However, a significant difference was obtained on the rate of early cleavage and useful blastulation when our cohort was compared to mature in vivo oocytes from different patients (control group).

Conclusion: This study has shown that after incubation in vitro maturation autologous cumulus cell co-culture with cumulus-oocyte cells, immature oocytes recovered during stimulated cycles can give rise to competent oocytes, i.e., capable of being fertilized, of cleaving, and of developing into embryos up to the blastocyst stage. Our study therefore seems to be in the direction of a favorable use of these immature oocytes obtained after stimulated IVF-ICSI cycles. The continuation of this study by including a larger number of oocytes is necessary in order to evaluate the real contribution of this technique in routine.

Inhibin B

There is insufficient evidence to permit conclusions regarding the use of inhibin B as a measure of ovarian reserve. More studies are needed to support improved outcomes (i.e., increased successful pregnancies with delivery of liveborn children) with the use of this test.

The higher level of serum inhibin B of reproductive age women is one of the important factors to maintain a low level of serum FSH. However, with the increase of their age, both the quality and quantity of ovarian follicles decrease, the level of serum inhibin B decreases gradually, and the inhibitory effect on FSH will be weakened, which is also one of the important reasons for the progressive increase of their serum FSH levels. Inhibin B may have certain clinical application potential in assessing the progress of ovarian aging, diagnosing premature ovarian failure (POF) or premature ovarian insufficiency (POI), evaluating the ovarian function of cancer survivors, and predicting assisted reproductive technology (ART) outcomes (Wen, J. et. al. 2021).

Wen, J. et.al. (2021) conducted a study aimed to define the variation trend of inhibin B in healthy women with age and explore its value in the reflection of ovarian reserve. Although the results showed a slight advantage in predicting ovarian response, more studies are needed to provide clarity of the reference range and potential value of inhibin B.